P2 Chapter 9: Differentiation

Chapter Practice

Key Points

1 For small angles, measured in radians:

- $\sin x \approx x$
- $\cos x \approx 1 - \frac{1}{2}x^2$

2
- If $y = \sin kx$, then $\dfrac{dy}{dx} = k\cos kx$
- If $y = \cos kx$, then $\dfrac{dy}{dx} = -k\sin kx$

3
- If $y = e^{kx}$, then $\dfrac{dy}{dx} = ke^{kx}$
- If $y = \ln x$, then $\dfrac{dy}{dx} = \dfrac{1}{x}$

4 If $y = a^{kx}$, where k is a real constant and $a > 0$, then $\dfrac{dy}{dx} = a^{kx}k\ln a$

5 The **chain rule** is: $\dfrac{dy}{dx} = \dfrac{dy}{du} \times \dfrac{du}{dx}$

where y is a function of u and u is another function of x.

6 The chain rule enables you to differentiate a function of a function. In general,

- if $y = (f(x))^n$ then $\dfrac{dy}{dx} = n(f(x))^{n-1}f'(x)$
- if $y = f(g(x))$ then $\dfrac{dy}{dx} = f'(g(x))g'(x)$

Key Points

7 $\dfrac{dy}{dx} = \dfrac{1}{\dfrac{dx}{dy}}$

8 The **product rule**:

- If $y = uv$ then $\dfrac{dy}{dx} = u\dfrac{dv}{dx} + v\dfrac{du}{dx}$, where u and v are functions of x.
- If $f(x) = g(x)h(x)$ then $f'(x) = g(x)h'(x) + h(x)g'(x)$

9 The **quotient rule**:

- If $y = \dfrac{u}{v}$, then $\dfrac{dy}{dx} = \dfrac{v\dfrac{du}{dx} - u\dfrac{dv}{dx}}{v^2}$ where u and v are functions of x.
- If $f(x) = \dfrac{g(x)}{h(x)}$, then $f'(x) = \dfrac{h(x)g'(x) - g(x)h'(x)}{(h(x))^2}$

10

- If $y = \tan kx$, then $\dfrac{dy}{dx} = k\sec^2 kx$
- If $y = \operatorname{cosec} kx$, then $\dfrac{dy}{dx} = -k\operatorname{cosec} kx \cot kx$
- If $y = \sec kx$, then $\dfrac{dy}{dx} = k\sec kx \tan kx$
- If $y = \cot kx$, then $\dfrac{dy}{dx} = -k\operatorname{cosec}^2 kx$

Key Points

11 If x and y are given as functions of a parameter, t: $\dfrac{dy}{dx} = \dfrac{\frac{dy}{dt}}{\frac{dx}{dt}}$

12
- $\dfrac{d}{dx}(f(y)) = f'(y)\dfrac{dy}{dx}$
- $\dfrac{d}{dx}(y^n) = ny^{n-1}\dfrac{dy}{dx}$
- $\dfrac{d}{dx}(xy) = x\dfrac{dy}{dx} + y$

13
- The function $f(x)$ is **concave** on a given interval if and only if $f''(x) \leqslant 0$ for every value of x in that interval.
- The function $f(x)$ is **convex** on a given interval if and only if $f''(x) \geqslant 0$ for every value of x in that interval.

14 A **point of inflection** is a point at which $f''(x)$ changes sign.

15 You can use the chain rule to connect rates of change in situations involving more than two variables.

Chapter Exercises

1 Differentiate with respect to x:

a $\ln x^2$ **(3 marks)**

b $x^2 \sin 3x$ **(4 marks)**

2 **a** Given that $2y = x - \sin x \cos x$, $0 < x < 2\pi$, show that $\frac{dy}{dx} = \sin^2 x$. **(4 marks)**

b Find the coordinates of the points of inflection of the curve. **(4 marks)**

3 Differentiate, with respect to x:

a $\frac{\sin x}{x}$, $x > 0$ **(4 marks)**

b $\ln \frac{1}{x^2 + 9}$ **(4 marks)**

4 $f(x) = \frac{x}{x^2 + 2}$, $x \in \mathbb{R}$

a Given that $f(x)$ is increasing on the interval $[-k, k]$, find the largest possible value of k. **(4 marks)**

b Find the exact coordinates of the points of inflection of $f(x)$. **(5 marks)**

5 The function f is defined for positive real values of x by

$$f(x) = 12 \ln x + x^{\frac{3}{2}}$$

a Find the set of values of x for which $f(x)$ is an increasing function of x. **(4 marks)**

b Find the coordinates of the point of inflection of the function f. **(4 marks)**

Chapter Exercises

6 Given that a curve has equation $y = \cos^2 x + \sin x$, $0 < x < 2\pi$, find the coordinates of the stationary points of the curve. **(6 marks)**

7 The maximum point on the curve with equation $y = x\sqrt{\sin x}$, $0 < x < \pi$, is the point A. Show that the x-coordinate of point A satisfies the equation $2\tan x + x = 0$. **(5 marks)**

8 $f(x) = e^{0.5x} - x^2$, $x \in \mathbb{R}$

a Find $f'(x)$. **(3 marks)**

b By evaluating $f'(6)$ and $f'(7)$, show that the curve with equation $y = f(x)$ has a stationary point at $x = p$, where $6 < p < 7$. **(2 marks)**

9 $f(x) = e^{2x} \sin 2x$, $0 < x < \pi$

a Use calculus to find the coordinates of the turning points on the graph of $y = f(x)$. **(6 marks)**

b Show that $f''(x) = 8e^{2x} \cos 2x$. **(4 marks)**

c Hence, or otherwise, determine which turning point is a maximum and which is a minimum. **(3 marks)**

d Find the points of inflection of $f(x)$. **(2 marks)**

10 The curve C has equation $y = 2e^x + 3x^2 + 2$. Find the equation of the normal to C at the point where the curve intercepts the y-axis. Give your answer in the form $ax + by + c = 0$ where a, b and c are integers to be found. **(5 marks)**

Chapter Exercises

11 The curve C has equation $y = \mathrm{f}(x)$, where

$$\mathrm{f}(x) = 3\ln x + \frac{1}{x},\ x > 0$$

The point P is a stationary point on C.

a Calculate the x-coordinate of P. **(4 marks)**

The point Q on C has x-coordinate 1.

b Find an equation for the normal to C at Q. **(4 marks)**

12 The curve C has equation $y = \mathrm{e}^{2x}\cos x$.

a Show that the turning points on C occur when $\tan x = 2$. **(4 marks)**

b Find an equation of the tangent to C at the point where $x = 0$. **(4 marks)**

13 Given that $x = y^2 \ln y$, $y > 0$,

a find $\dfrac{\mathrm{d}x}{\mathrm{d}y}$ **(4 marks)**

b Use your answer to part **a** to find in terms of e, the value of $\dfrac{\mathrm{d}y}{\mathrm{d}x}$ at $y = \mathrm{e}$. **(2 marks)**

14 A curve has equation $\mathrm{f}(x) = (x^3 - 2x)\mathrm{e}^{-x}$.

a Find $\mathrm{f}'(x)$. **(4 marks)**

The normal to C at the origin O intersects C again at P.

b Show that the x-coordinate of P is the solution to the equation $2x^2 = \mathrm{e}^x + 4$. **(6 marks)**

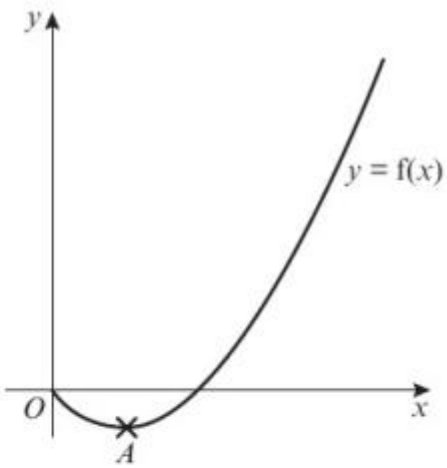
Chapter Exercises

15 The diagram shows part of the curve with equation $y = f(x)$ where $f(x) = x(1 + x)\ln x$, $x > 0$

The point A is the minimum point of the curve.

a Find $f'(x)$. **(4 marks)**

b Hence show that the x-coordinate of A is the solution to the equation $x = e^{-\frac{1+x}{1+2x}}$ **(4 marks)**



16 The curve C is given by the equations

$$x = 4t - 3, \quad y = \frac{8}{t^2}, \quad t > 0$$

where t is a parameter.

At A, $t = 2$. The line l is the normal to C at A.

a Find $\frac{dy}{dx}$ in terms of t. **(4 marks)**

b Hence find an equation of l. **(3 marks)**

17 The curve C is given by the equations $x = 2t$, $y = t^2$, where t is a parameter.
Find an equation of the normal to C at the point P on C where $t = 3$. **(7 marks)**

18 The curve C has parametric equations

$$x = t^3, \quad y = t^2, \quad t > 0$$

Find an equation of the tangent to C at A (1, 1). **(7 marks)**

Chapter Exercises

19 A curve C is given by the equations

$$x = 2\cos t + \sin 2t, \quad y = \cos t - 2\sin 2t, \quad 0 < t < \pi$$

where t is a parameter.

a Find $\frac{dx}{dt}$ and $\frac{dy}{dt}$ in terms of t. **(3 marks)**

b Find the value of $\frac{dy}{dx}$ at the point P on C where $t = \frac{\pi}{4}$ **(3 marks)**

c Find an equation of the normal to the curve at P. **(3 marks)**

20 A curve is given by $x = 2t + 3$, $y = t^3 - 4t$, where t is a parameter. The point A has parameter $t = -1$ and the line l is the tangent to C at A. The line l also cuts the curve at B.

a Show that an equation for l is $2y + x = 7$. **(6 marks)**

b Find the value of t at B. **(5 marks)**

21 A car has value £V at time t years. A model for V assumes that the rate of decrease of V at time t is proportional to V. Form an appropriate differential equation for V.

22 In a study of the water loss of picked leaves the mass, M grams, of a single leaf was measured at times, t days, after the leaf was picked. It was found that the rate of loss of mass was proportional to the mass M of the leaf.
Write down a differential equation for the rate of change of mass of the leaf.

Chapter Exercises

23 In a pond the amount of pondweed, P, grows at a rate proportional to the amount of pondweed already present in the pond. Pondweed is also removed by fish eating it at a constant rate of Q per unit of time.
Write down a differential equation relating P to t, where t is the time which has elapsed since the start of the observation.

24 A circular patch of oil on the surface of some water has radius r and the radius increases over time at a rate inversely proportional to the radius.
Write down a differential equation relating r and t, where t is the time which has elapsed since the start of the observation.

25 A metal bar is heated to a certain temperature, then allowed to cool down and it is noted that, at time t, the rate of loss of temperature is proportional to the difference between the temperature of the metal bar, θ, and the temperature of its surroundings θ_0.
Write down a differential equation relating θ and t.

26 The curve C has parametric equations

$$x = 4\cos 2t, \quad y = 3\sin t, \quad -\frac{\pi}{2} < t < \frac{\pi}{2}$$

A is the point $\left(2, \frac{3}{2}\right)$, and lies on C.

a Find the value of t at the point A. **(2 marks)**

b Find $\frac{dy}{dx}$ in terms of t. **(3 marks)**

c Show that an equation of the normal to C at A is $6y - 16x + 23 = 0$. **(4 marks)**

The normal at A cuts C again at the point B.

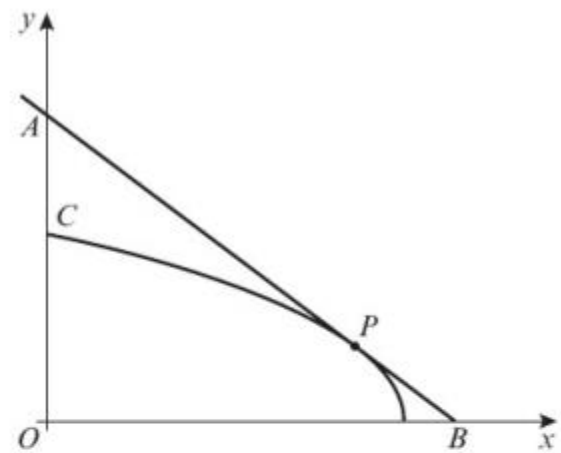
d Find the y-coordinate of the point B. **(6 marks)**

Chapter Exercises

27 The diagram shows the curve C with parametric equations

$$x = a\sin^2 t, \quad y = a\cos t, \quad 0 \leqslant t \leqslant \tfrac{1}{2}\pi$$

where a is a positive constant. The point P lies on C and has coordinates $\left(\frac{3}{4}a, \frac{1}{2}a\right)$.



a Find $\dfrac{dy}{dx}$, giving your answer in terms of t. **(4 marks)**

b Find an equation of the tangent to C at P. **(4 marks)**

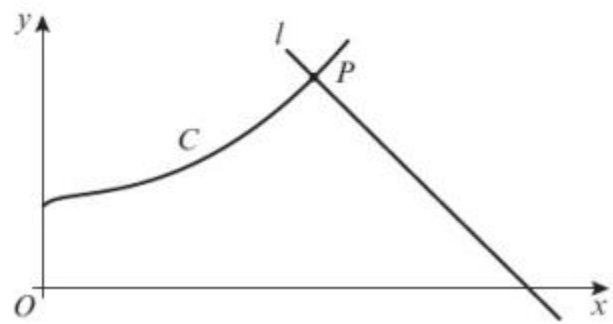
The tangent to C at P cuts the coordinate axes at points A and B.

c Show that the triangle AOB has area ka^2 where k is a constant to be found. **(2 marks)**

28 This graph shows part of the curve C with parametric equations

$$x = (t+1)^2, \quad y = \tfrac{1}{2}t^3 + 3, \quad t > -1$$

P is the point on the curve where $t = 2$.
The line l is the normal to C at P.



Find the equation of l. **(7 marks)**

Chapter Exercises

29 Find the gradient of the curve with equation $5x^2 + 5y^2 - 6xy = 13$ at the point (1, 2). **(7 marks)**

30 Given that $e^{2x} + e^{2y} = xy$, find $\frac{dy}{dx}$ in terms of x and y. **(7 marks)**

31 Find the coordinates of the turning points on the curve $y^3 + 3xy^2 - x^3 = 3$. **(7 marks)**

32 **a** If $(1 + x)(2 + y) = x^2 + y^2$, find $\frac{dy}{dx}$ in terms of x and y. **(4 marks)**

b Find the gradient of the curve $(1 + x)(2 + y) = x^2 + y^2$ at each of the two points where the curve meets the y-axis. **(3 marks)**

c Show also that there are two points at which the tangents to this curve are parallel to the y-axis. **(4 marks)**

33 A curve has equation $7x^2 + 48xy - 7y^2 + 75 = 0$. A and B are two distinct points on the curve and at each of these points the gradient of the curve is equal to $\frac{2}{11}$. Use implicit differentiation to show that the straight line passing through A and B has equation $x + 2y = 0$. **(6 marks)**

34 Given that $y = x^x$, $x > 0$, $y > 0$, by taking logarithms show that

$$\frac{dy}{dx} = x^x(1 + \ln x)$$ **(6 marks)**

Chapter Exercises

35 **a** Given that $a^x \equiv e^{kx}$, where a and k are constants, $a > 0$ and $x \in \mathbb{R}$, prove that $k = \ln a$. **(2 marks)**

b Hence, using the derivative of e^{kx}, prove that when $y = 2^x$

$$\frac{dy}{dx} = 2^x \ln 2$$ **(4 marks)**

c Hence deduce that the gradient of the curve with equation $y = 2^x$ at the point (2, 4) is ln 16. **(3 marks)**

36 A population P is growing at the rate of 9% each year and at time t years may be approximated by the formula

$$P = P_0(1.09)^t, \ t \geqslant 0$$

where P is regarded as a continuous function of t and P_0 is the population at time $t = 0$.

a Find an expression for t in terms of P and P_0. **(2 marks)**

b Find the time T years when the population has doubled from its value at $t = 0$, giving your answer to 3 significant figures. **(4 marks)**

c Find, as a multiple of P_0, the rate of change of population $\frac{dP}{dt}$ at time $t = T$. **(4 marks)**

37 A curve C has equation

$$y = \ln(\sin x), \quad 0 < x < \pi$$

a Find the stationary point of the curve C. **(6 marks)**

b Show that the curve C is concave at all values of x in its given domain. **(3 marks)**

38 The mass of a radioactive substance t years after first being observed is modelled by the equation

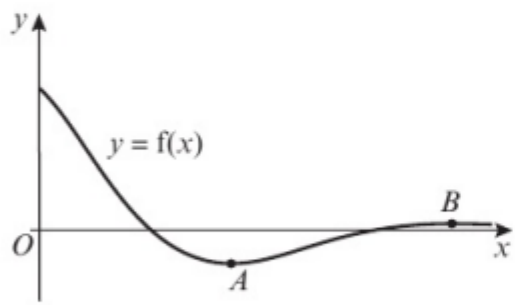
$$m = 40e^{-0.244t}$$

a Find the mass of the substance nine months after it was first observed. **(2 marks)**

b Find $\frac{dm}{dt}$ **(2 marks)**

c With reference to the model, interpret the significance of the sign of the value of $\frac{dm}{dt}$ found in part **b**. **(1 mark)**

39 The curve C with equation $y = f(x)$ is shown in the diagram, where $f(x) = \frac{\cos 2x}{e^x}$, $0 \leqslant x \leqslant \pi$



The curve has a local minimum at A and a local maximum at B.

a Show that the x-coordinates of A and B satisfy the equation $\tan 2x = -0.5$ and hence find the coordinates of A and B. **(6 marks)**

b Using your answer to part **a**, find the coordinates of the maximum and minimum turning points on the curve with equation $y = 2 + 4f(x - 4)$. **(3 marks)**

c Determine the values of x for which $f(x)$ is concave. **(5 marks)**

Chapter Exercises

Challenge

The curve C has parametric equations

$$y = 2\sin 2t, \quad x = 5\cos\left(t + \frac{\pi}{12}\right), \quad 0 \leqslant t \leqslant 2\pi$$

a Find $\frac{dy}{dx}$ in terms of t.

b Find the coordinates of the points on C where $\frac{dy}{dx} = 0$.

c Find the coordinates of any points where the curve cuts or intersects the coordinate axes, and determine the gradient of the curve at these points.

d Find the coordinates of the points on C where $\frac{dx}{dy} = 0$.

e Hence sketch C.

Hint The points on C where $\frac{dx}{dy} = 0$ correspond to points where a tangent to the curve would be a vertical line.

Chapter Answers

1 **a** $\frac{2}{x}$ **b** $2x\sin 3x + 3x^2\cos 3x$

2 **a** $2\frac{dy}{dx} = 1 - \sin x\frac{d}{dx}(\cos x) - \frac{d}{dx}(\sin x)\cos x$
$= 1 + \sin^2 x - \cos^2 x = 2\sin^2 x$
So $\frac{dy}{dx} = \sin^2 x$
b $\left(\frac{\pi}{2}, \frac{\pi}{4}\right), \left(\pi, \frac{\pi}{2}\right), \left(\frac{3\pi}{2}, \frac{3\pi}{4}\right)$

3 **a** $\frac{x\cos x - \sin x}{x^2}$ **b** $-\frac{2x}{x^2+9}$

4 **a** $k = \sqrt{2}$ **b** $(0, 0), \left(\pm\sqrt{6}, \pm\frac{\sqrt{3}}{4\sqrt{2}}\right)$

5 **a** $x > 0$ **b** $\left(\sqrt[3]{256}, 32\ln 2 + 16\right)$

6 $\left(\frac{\pi}{6}, \frac{5}{4}\right), \left(\frac{\pi}{2}, 1\right), \left(\frac{5\pi}{6}, \frac{5}{4}\right), \left(\frac{3\pi}{2}, -1\right)$

7 Maximum is when $\frac{dy}{dx} = 0$
$\frac{dy}{dx} = \sqrt{\sin x} + x\left(\cos x \times \frac{1}{2\sqrt{\sin x}}\right) = \frac{2\sin x + x\cos x}{2\sqrt{\sin x}} = 0$
So $2\sin x + x\cos x = 0 \Rightarrow 2\sin x = -x\cos x \Rightarrow 2\tan x = -x$
$\therefore 2\tan x + x = 0$

8 **a** $f'(x) = 0.5e^{0.5x} - 2x$
b $f'(6) = -1.957\ldots < 0$, $f'(7) = 2.557\ldots > 0$
So there exists $p \in (6, 7)$ such that $f'(p) = 0$.
$\therefore$ there is a stationary point for some $x = p \in (6, 7)$.

9 **a** $\left(\frac{3\pi}{8}, \frac{e^{\frac{3\pi}{4}}}{\sqrt{2}}\right), \left(\frac{7\pi}{8}, -\frac{e^{\frac{7\pi}{4}}}{\sqrt{2}}\right)$
b $f''(x) = 2e^{2x}(-2\sin 2x + 2\cos 2x) + 4e^{2x}(\cos 2x + \sin 2x)$
$= 4e^{2x}(-\sin 2x + \cos 2x + \cos 2x + \sin 2x)$
$= 8e^{2x}\cos 2x$
c $\left(\frac{3\pi}{8}, \frac{e^{\frac{3\pi}{4}}}{\sqrt{2}}\right)$ is a maximum; $\left(\frac{7\pi}{8}, -\frac{e^{\frac{7\pi}{4}}}{\sqrt{2}}\right)$ is a minimum.
d $\left(\frac{\pi}{4}, e^{\frac{\pi}{2}}\right), \left(\frac{3\pi}{4}, -e^{\frac{3\pi}{2}}\right)$

10 $x + 2y - 8 = 0$

11 **a** $x = \frac{1}{3}$ **b** $y = -\frac{1}{2}x + 1\frac{1}{2}$

12 **a** $f'(x) = e^{2x}(2\cos x - \sin x)$ **b** $y = 2x + 1$
$2\cos x - \sin x = 0 \Rightarrow \tan x = 2$

13 **a** $y + 2y\ln y$ **b** $\frac{1}{3e}$

14 **a** $e^{-x}(-x^3 + 3x^2 + 2x - 2)$
b $f'(0) = -2 \Rightarrow$ gradient of normal $= \frac{1}{2}$
Equation of normal is $y = \frac{1}{2}x$
$(x^3 - 2x)e^{-x} = \frac{1}{2}x \Rightarrow 2x^3 - 4x = xe^x \Rightarrow 2x^2 = e^x + 4$

15 **a** $1 + x + (1 + 2x)\ln x$
b $1 + x + (1 + 2x)\ln x = 0 \Rightarrow x = e^{-\frac{1+x}{1+2x}}$

16 **a** $\frac{dy}{dx} = -\frac{4}{t^3}$ **b** $y = 2x - 8$

17 $3y + x = 33$ **18** $y = \frac{2}{3}x + \frac{1}{3}$

19 **a** $\frac{dx}{dt} = -2\sin t + 2\cos 2t$; $\frac{dy}{dt} = -\sin t - 4\cos 2t$

b $\frac{1}{2}$ **c** $y + 2x = \frac{5\sqrt{2}}{2}$

20 **a** $\frac{dy}{dt} = 3t^2 - 4$, $\frac{dx}{dt} = 2$, $\frac{dy}{dx} = \frac{3t^2 - 4}{2}$

At $t = -1$, $\frac{dy}{dx} = -\frac{1}{2}$, $x = 1$, $y = 3$.

Equation of l is $2y + x = 7$.

b 2

21 $\frac{dV}{dt} = -kV$ **22** $\frac{dM}{dt} = -kM$ **23** $\frac{dP}{dt} = kP - Q$

24 $\frac{dr}{dt} = \frac{k}{r}$ **25** $\frac{d\theta}{dt} = -k(\theta - \theta_0)$

26 **a** $\frac{\pi}{6}$ **b** $-\frac{3}{16}\operatorname{cosec} t$

c $\frac{dy}{dx} = -\frac{3}{8} \Rightarrow$ gradient of normal $= \frac{8}{3}$

$y - \frac{3}{2} = \frac{8}{3}(x - 2) \Rightarrow 6y - 16x + 23 = 0$

d $-\frac{123}{64}$

27 **a** $-\frac{1}{2}\sec t$ **b** $4y + 4x = 5a$

c Tangent crosses the x-axis at $x = \frac{5}{4}a$, and crosses the y-axis at $y = \frac{5}{4}a$.

So area $AOB = \frac{1}{2}\left(\frac{5}{4}a\right)^2 = \frac{25}{32}a^2$, $k = \frac{25}{32}$

28 $y + x = 16$

29 $\frac{1}{7}$

30 $\frac{y - 2e^{2x}}{2e^{2y} - x}$ **31** $(1, 1)$ and $(-\sqrt[3]{-3}, \sqrt[3]{-3})$.

32 **a** $\frac{2x - 2 - y}{1 + x - 2y}$ **b** $\frac{4}{3}, -\frac{1}{3}$

c $\left(\frac{5 + 2\sqrt{13}}{3}, \frac{4 + \sqrt{13}}{3}\right)$ and $\left(\frac{5 - 2\sqrt{13}}{3}, \frac{4 - \sqrt{13}}{3}\right)$

33 $14x + 48y + 48x\frac{dy}{dx} - 14y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-7x - 24y}{24x - 7y}$

So $\frac{-7x - 24y}{24x - 7y} = \frac{2}{11} \Rightarrow -77x - 264y$

$= 48x - 14y \Rightarrow x + 2y = 0$

34 $\ln y = x\ln x \Rightarrow \frac{1}{y} \times \frac{dy}{dx} = x\frac{d}{dx}(\ln x) + \frac{d}{dx}(x)\ln x = 1 + \ln x$

So $\frac{dy}{dx} = y(1 + \ln x) = x^x(1 + \ln x)$

35 **a** $\ln a^x = \ln e^{kx} \Rightarrow x\ln a = kx\ln e = kx \Rightarrow k = \ln a$

b $y = e^{(\ln 2)x} \Rightarrow \frac{dy}{dx} = \ln 2\, e^{(\ln 2)x} = 2^x \ln 2$

c $\frac{dy}{dx} = 2^2 \ln 2 = 4\ln 2 = \ln 2^4 = \ln 16$

36 **a** $\frac{\ln P - \ln P_0}{\ln 1.09}$ **b** 8.04 years **c** $0.172P_0$

37 **a** $\left(\frac{\pi}{2}, 0\right)$

b $\frac{d^2y}{dx^2} = -\text{cosec}^2 x$. $\text{cosec}^2 x > 0$ for all x,

hence $-\text{cosec}^2 x < 0$, so $\frac{d^2y}{dx^2} < 0$ for all x.

Thus C is concave for all values of x.

38 **a** $40e^{-0.183} = 33.31\ldots$ **b** $-9.76e^{-0.244t}$

c The mass is decreasing

39 **a** $f'(x) = -\frac{2\sin 2x + \cos 2x}{e^x}$

$f'(x) = 0 \Leftrightarrow 2\sin 2x + \cos 2x = 0 \Leftrightarrow \tan 2x = -0.5$

$A(1.34, -0.234)$, $B(2.91, 0.0487)$

b Maximum (6.91, 2.19); minimum (5.34, 1.06) to 3 s.f.

c $0 < x \leqslant 0.322$, $1.89 \leqslant x < \pi$

Challenge

a $-\frac{4\cos 2t}{5\sin\left(t + \frac{\pi}{12}\right)}$

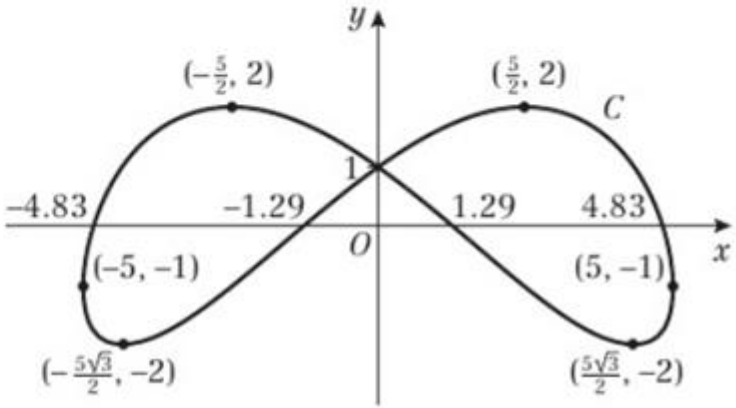
b $\left(\frac{5}{2}, 2\right), \left(-\frac{5\sqrt{3}}{2}, -2\right), \left(-\frac{5}{2}, 2\right), \left(\frac{5\sqrt{3}}{2}, -2\right)$

c Cuts the x–axis at:

(4.83, 0) gradient −3.09; (−1.29, 0) gradient 0.828

(−4.83, 0) gradient 3.09; (1.29, 0) gradient −0.828

Cuts the y–axis twice at (0, 1) gradients 0.693 and −0.693

d (−5, −1) and (5, −1)

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